```
513         if not self.enabled:
514             return []
515         limit = max(1, int(limit))
516         cur = self._conn.cursor()
517         cur.execute(
518             """
519             SELECT
520                 flight_key,
521                 icao_address,
522                 callsign,
523                 timestamp,
524                 lat_e6,
525                 lon_e6,
526                 altitude_m,
527                 speed_cms,
528                 heading_de2,
529                 vs_cms,
530                 squawk,
531                 source_ip,
532                 sample_count
533             FROM flight_latest
534             ORDER BY timestamp DESC
535             LIMIT ?
536             """,
537             (limit,),
538         )
539         rows = []
540         for r in cur.fetchall():
541             rows.append(
542                 {
543                     "flightKey": r["flight_key"],
544                     "icaoAddress": r["icao_address"],
545                     "callsign": r["callsign"],
546                     "timestamp": r["timestamp"],
547                     "latDD": r["lat_e6"] / 1000000.0,
548                     "lonDD": r["lon_e6"] / 1000000.0,
549                     "altitudeM": r["altitude_m"],
550                     "speedCMS": r["speed_cms"],
551                     "headingDE2": r["heading_de2"],
552                     "verVelocityCMS": r["vs_cms"],
553                     "squawk": r["squawk"],
554                     "sourceIp": r["source_ip"],
555                     "sampleCount": r["sample_count"],
556                 }
557             )
558         return rows
559 
560     def write_latest_matrix_csv(self, output_path, limit=5000):
561         if not self.enabled:
562             return None
563 
564         rows = self.get_latest_matrix(limit=limit)
565         out_path = Path(output_path)
566         out_path.parent.mkdir(parents=True, exist_ok=True)
567 
568         headers = [
569             "flightKey",
570             "icaoAddress",
571             "callsign",
572             "timestamp",
573             "latDD",
574             "lonDD",
575             "altitudeM",
576             "speedCMS",
577             "headingDE2",
578             "verVelocityCMS",
579             "squawk",
580             "sourceIp",
581             "sampleCount",
582         ]
583 
584         try:
585             with out_path.open("w", newline="", encoding="utf-8") as f:
```